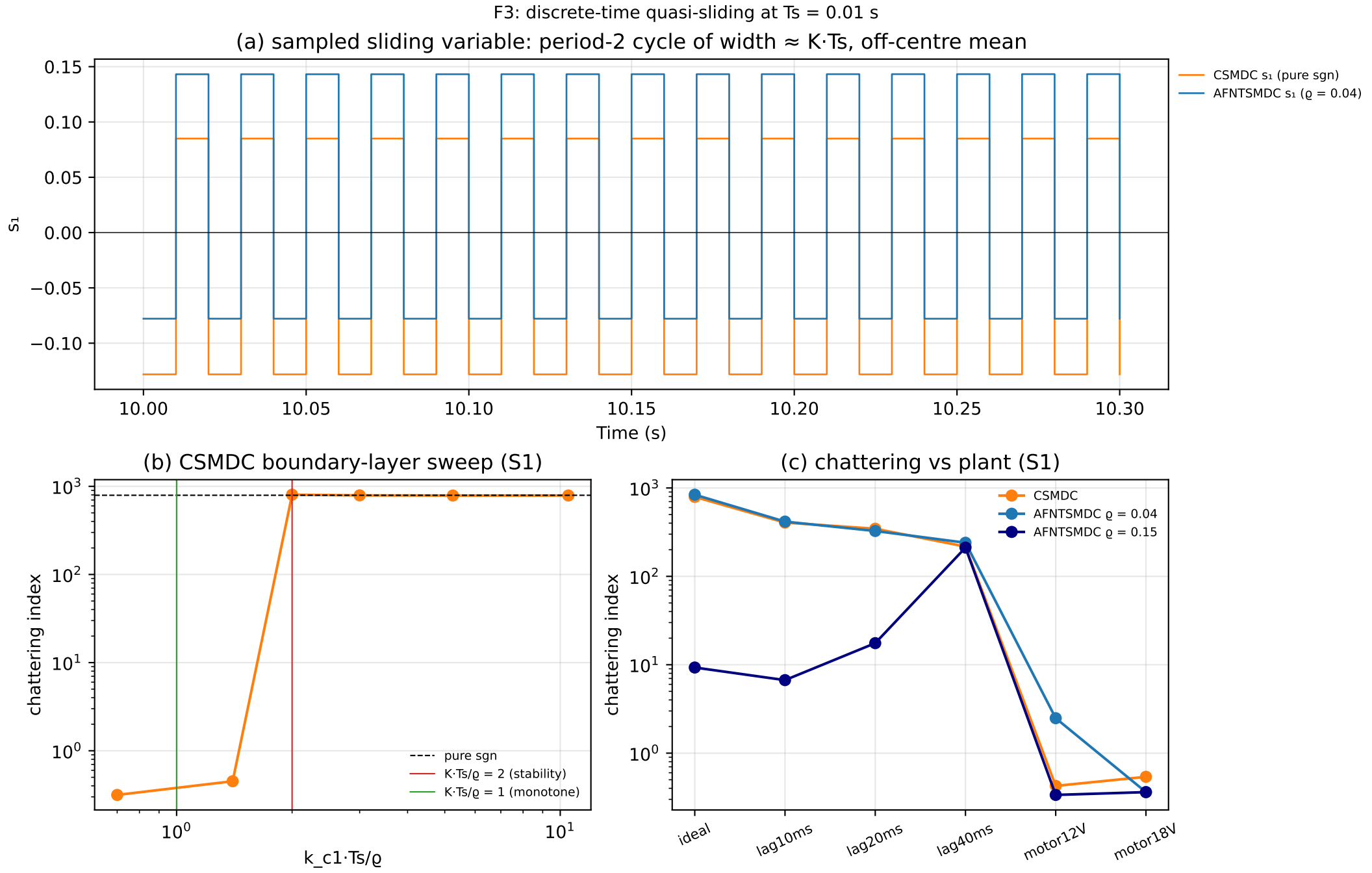
F3: discrete-time quasi-sliding at Ts = 0.01 s
(a) sampled sliding variable: period-2 cycle of width ≈ K·Ts, off-centre mean
s₁
Time (s)
CSMDC s₁ (pure sgn)
AFNTSMDC s₁ (ϱ = 0.04)
(b) CSMDC boundary-layer sweep (S1)
chattering index
k_c1·Ts/ϱ
pure sgn
K·Ts/ϱ = 2 (stability)
K·Ts/ϱ = 1 (monotone)
(c) chattering vs plant (S1)
CSMDC
AFNTSMDC ϱ = 0.04
AFNTSMDC ϱ = 0.15
ideal
lag10ms
lag20ms
lag40ms
motor12V
motor18V